Reproduction of the Dehbi (2008) CRW–Langevin Particle Dispersion Model in a Turbulent 90° Pipe Bend Using ANSYS Fluent and a User-Defined Function

Technical report, revision 2

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Revision note

Revision 1 of this report (March 2026) is kept unchanged in `original/report-dehbi-v1-2026-03.pdf`. Revision 2 keeps the same simulations, settings and results. It changes the presentation in four ways:

1. The model equations are written out as they are implemented in `udf/crw_dehbi2008_rev10.c`, with source line references.
2. The deposition fractions are tabulated from the Fluent DPM summaries that were already reproduced in Appendix B of revision 1.
3. Statements of revision 1 that the evidence does not support (for example “excellent agreement” and “true mesh independence”) are replaced by what was actually measured.
4. A code review of the UDF is added (Section 7). It identifies two implementation issues that affect the physics. They are reported here, not corrected, because the reported results were produced with the code as it stands.

No simulation was re-run for this revision.

1. Objective

The study reproduces the 90° pipe-bend deposition case of Dehbi (2008), who coupled a RANS flow solution with a continuous random walk (CRW) model based on a normalised Langevin equation. Standard eddy-interaction (discrete random walk, DRW) models in RANS solvers assume locally homogeneous, isotropic turbulence. In wall-bounded flows they violate the well-mixed condition: particles with low inertia drift towards regions of low turbulence intensity and accumulate artificially near the wall, which over-predicts deposition. Dehbi’s model counters this with anisotropic near-wall velocity statistics and a drift term proportional to the wall-normal gradient of the velocity variance.

The specific objectives were:

1. Compute the turbulent flow in the bend with the Reynolds stress model (RSM).
2. Implement the CRW–Langevin model as a compiled ANSYS Fluent UDF that replaces the built-in DRW dispersion model.
3. Compute the deposition fraction for particle diameters of 5 to 50 µm and compare it with Dehbi (2008, Fig. 8) and the experiments of Pui et al. (1987).

2. Geometry and flow conditions

Quantity	Value	Source
Pipe diameter D	0.02 m	revision 1, Sec. 2
Bend curvature ratio R/r	4 (R = 0.04 m)	revision 1, Sec. 2
Straight inlet / outlet sections	$5D$ (0.10 m) each	revision 1, Sec. 2
Fluid	air, ρ = 1.225 kg m^{-3}, μ = 1.81 × 10^{-5} Pa s	revision 1, Sec. 4.1
Inlet	uniform velocity, U = 3.0 m s^{-1}	revision 1, Sec. 4.1
Reynolds number $Re = \rho UD/\mu$	4.06 × 10^3 (computed from the values above; revision 1 states 4080)	computed
Outlet	pressure outlet, 0 Pa gauge	revision 1, Sec. 4.1
Walls	stationary, no slip	revision 1, Sec. 4.1



Figure 1. Computational domain: 90° bend with straight inlet and outlet sections.

Open item. Revision 1 states that these conditions replicate Pui et al. (1987). The geometry and Reynolds number have not been cross-checked against the case definition in Dehbi (2008), Sec. 10, for this revision. Because the comparison in Section 6 is made at equal particle diameter, any difference in D, U or R/r from the benchmark case changes the Stokes number at a given diameter and must be resolved before the comparison is considered final.

3. Mesh

The domain was meshed with structured hexahedral cells. The first cell height is 70 µm, which gives a wall y^+ ≈ 0.95 (Figure 3). The near-wall region contains 30 inflation layers with a growth ratio of 1.1. The final mesh has about 5.3 × 10^6 cells; a baseline mesh of about 2.5 × 10^6 cells was used first.

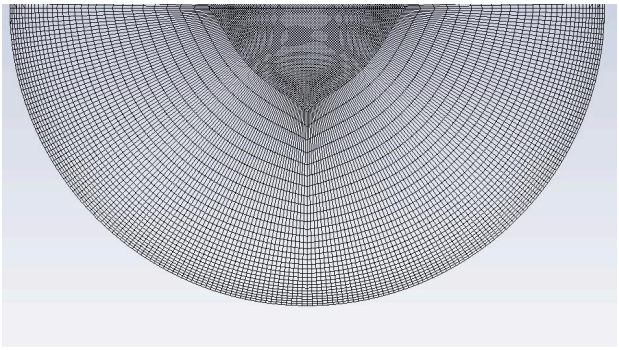
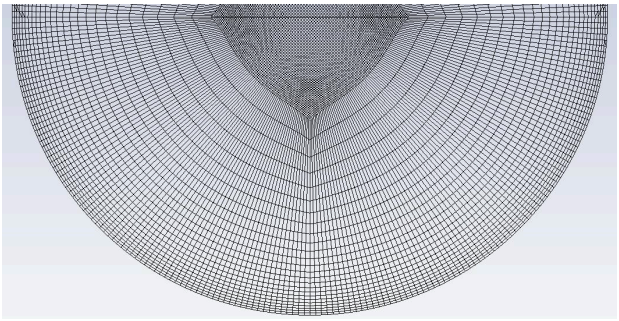
Refined mesh, 5.3 × 10^6 cells	Baseline mesh, 2.5 × 10^6 cells
	

Figure 2. Cross-section of the bend. The refined mesh adds resolution in the bend and in the region $y^+ < 100$.

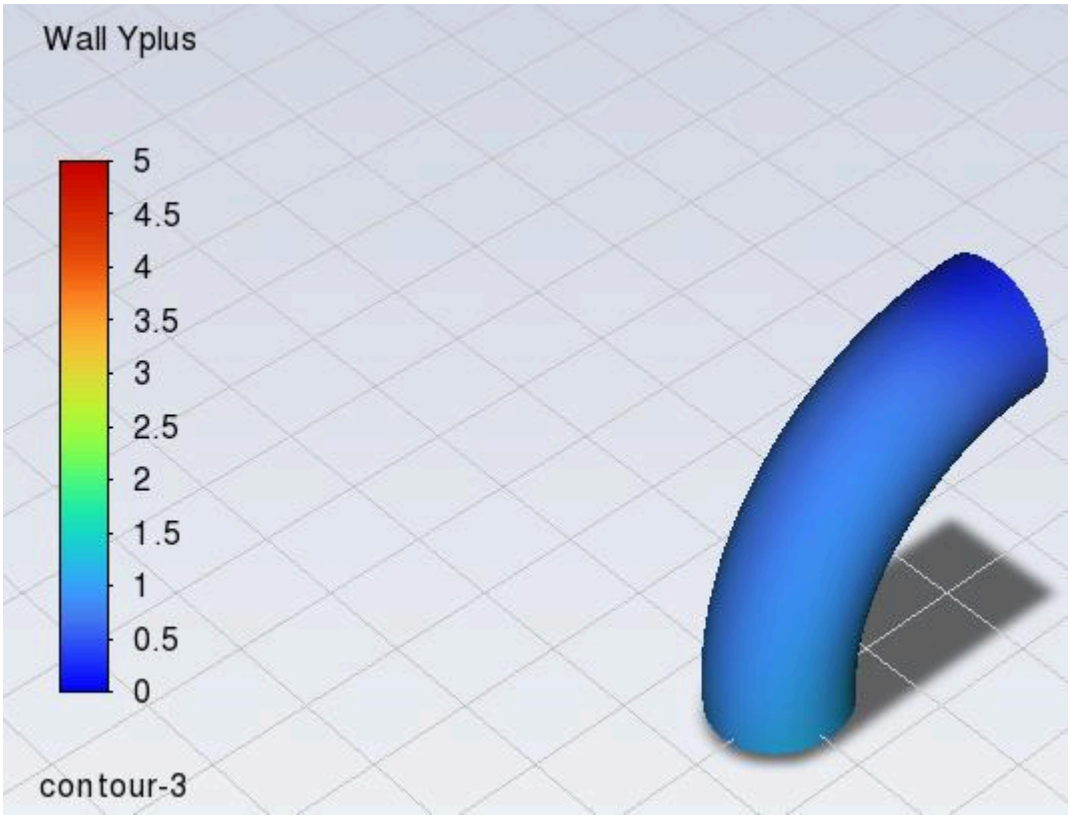


Figure 3. Wall y^+ on the bend surface (colour scale 0 to 5).

Mesh quality histograms (orthogonal quality and skewness) are in `figures/mesh-orthogonal-quality.png` and `figures/mesh-skewness.png`.

Mesh sensitivity. Revision 1 reports that the baseline mesh gave deposition fractions for 5 and 10 µm particles that deviated from the benchmark, while larger particles were insensitive, and that the refined mesh removed the deviation. The deposition fractions obtained on the baseline mesh were not recorded in revision 1. The study therefore demonstrates mesh sensitivity for small particles but does not quantify grid convergence; the phrase “true mesh independence” used in revision 1 is withdrawn.

4. Solver settings

Item	Setting
Solver	ANSYS Fluent 2024 R2, pressure based, steady
Turbulence model	Reynolds stress model, linear pressure-strain, enhanced wall treatment (Figure S1)
Pressure–velocity coupling	SIMPLE
Spatial discretisation	least-squares gradients; second-order pressure; second-order upwind for momentum, k, ε and Reynolds stresses (Figure S2)
Convergence	scaled residuals of all equations levelled off between about 10^{-3} and 10^{-6} after 1000 iterations (Figure 4)
Discrete phase	inert spheres, one-way coupling, steady tracking
Injection	surface injection at the inlet, 10,000 particles per diameter, randomised starting points
Particle density	1000 kg m^{-3} (revision 1, Sec. 4.3)
DPM tracking	accuracy control (tolerance 10^{-5}), automated high-/low-order scheme (Runge–Kutta / implicit) (Figure S4)
Built-in turbulent dispersion	disabled; dispersion supplied by the UDF body force
UDF hooks	body force `crw_langevin_force`, time step `crw_adaptive_dt`, 8 particle user scalars (Figure S3); wall boundary condition `dehbi_wall_bc`; on-demand `init_wall_udm`
Deposition criterion	particle trapped at first wall contact; deposition fraction = aborted / tracked

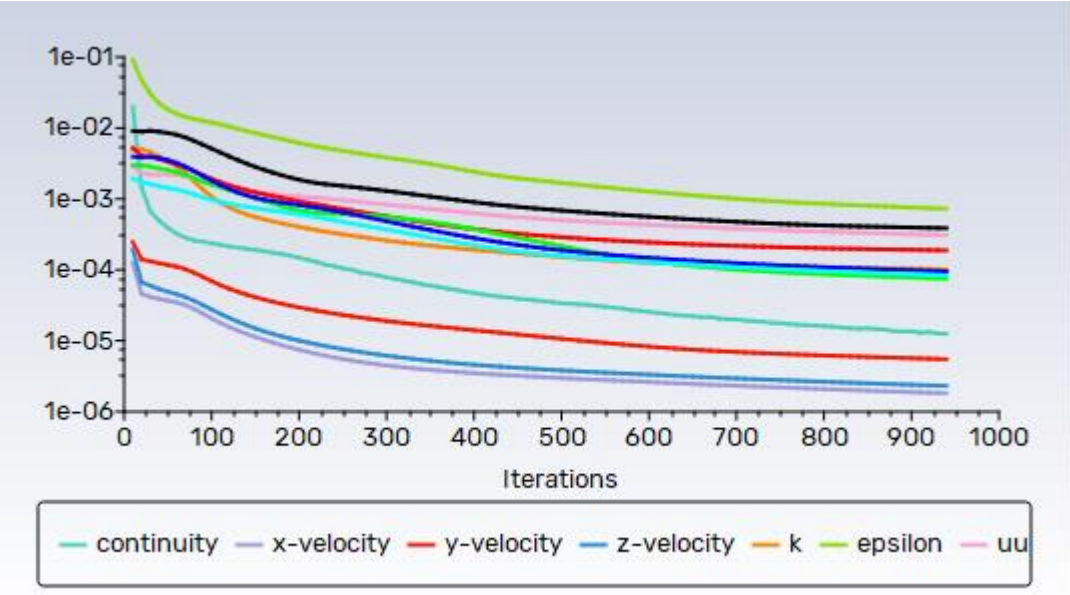


Figure 4. Scaled residuals of the continuous phase.

The Fluent dialogs reproduced in revision 1 are kept as supplementary figures in `figures/setup/`: S1 viscous model (`viscous-model.png`), S2 solution methods (`solution-methods.png`), S3 DPM UDF hooks (`dpm-udf-hooks.png`), S4 DPM numerics (`dpm-numerics.png`), S5 injection (`injection.png`) and S6 UDF compilation environment (`udf-compilation.png`).

Settings to confirm. The injection dialog in revision 1 (`figures/setup/injection.png`) shows three values that differ from the text and should be confirmed from the case file:

1. The injection named `particles-10micron` has a diameter of 5×10^{-6} m.
2. The particle material is `anthracite`, whose Fluent default density is 1550 kg m^{-3}, while the text states 1000 kg m^{-3}. The value used depends on whether the material density was edited; the dialog does not show it.
3. The particle injection velocity is 0 m s^{-1}, while the text and the UDF set-up notes state that particles are released at the gas velocity. With $\tau_p \le 7.7$ ms and a 0.10 m inlet section, particles relax to the gas velocity before the bend, so the effect on deposition is expected to be small, but the text should match the setting.

5. CRW–Langevin model as implemented

Symbols: u_τ friction velocity, ν kinematic viscosity, $y^+ = y u_\tau/\nu$, k turbulent kinetic energy, ε dissipation rate, τ_p particle relaxation time, τ_L fluid Lagrangian integral time, ξ standard normal deviate. Equation numbers in square brackets are those cited in the source code for Dehbi (2008).

5.1 Particle equation of motion

Fluent integrates the particle momentum equation with Schiller–Naumann drag computed from the mean slip velocity. The UDF adds the drag of the fluid velocity fluctuation as a body force per unit mass (lines 644–738):

$$\frac{F_i}{m_p} = f_D\, u'_i, \qquad f_D = \frac{1 + 0.15\, Re_p^{0.687}}{\tau_p}, \qquad \tau_p = \frac{\rho_p d_p^2}{18\,\mu}, \qquad Re_p = \frac{|\mathbf{U} - \mathbf{u}_p|\, d_p}{\nu}.$$

The fluctuation u'_i is updated once per particle time step, on the call for the first velocity component, and reused for the other two.

5.2 Normalised Langevin equation

The model evolves the normalised fluctuation $\eta_i = u'_i/\sigma_i$:

$$d\eta_i = -\frac{\eta_i}{\tau_L}\,dt + A_i\,dt + \sqrt{\frac{2}{\tau_L}}\,dW_i.$$

It is discretised semi-implicitly: the relaxation term is implicit, and the drift and noise terms are explicit (lines 397–399, 427–429):

$$\eta_i^{n+1} = \frac{\eta_i^n + \sqrt{2\,\Delta t/\tau_L}\;\xi_i + A_i\,\Delta t}{1 + \Delta t/\tau_L}.$$

The code comments describe this scheme as “implicit Euler”; it is more precisely a semi-implicit (linearly implicit) Euler–Maruyama scheme. For $\Delta t \ll \tau_L$, which holds here because $\Delta t \le 10^{-6}$ s, the stationary variance of η is $1/(1 + \Delta t/2\tau_L) \approx 1$.

5.3 Near-wall branch ($y^+ < 100$)

In the near-wall branch the equations are solved in a body-fitted frame (lines 225–275): $\mathbf{e}_2$ is the inward wall normal, $\mathbf{e}_1$ the local mean-velocity direction made orthogonal to $\mathbf{e}_2$, and $\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$. The root-mean-square fluctuations are the DNS fits attributed to Dreeben and Pope (1997) [Eqs. 37–39] (lines 131–148):

$$\frac{\sigma_1}{u_\tau} = \frac{0.40\, y^+}{1 + 0.0239\,(y^+)^{1.496}}, \qquad \frac{\sigma_2}{u_\tau} = \frac{0.0116\,(y^+)^2}{1 + 0.203\, y^+ + 0.00140\,(y^+)^{2.421}}, \qquad \frac{\sigma_3}{u_\tau} = \frac{0.19\, y^+}{1 + 0.0361\,(y^+)^{1.322}}.$$

The Lagrangian time scale follows Kallio and Reeks (1989) [Eqs. 31–33] (lines 196–214):

$$\tau_L = \tau_L^+ \frac{\nu}{u_\tau^2}, \qquad \tau_L^+ = \begin{cases} 10, & y^+ \le 5, \\ 7.122 + 0.5731\, y^+ - 0.00129\,(y^+)^2, & 5 < y^+ \le 100. \end{cases}$$

Only the wall-normal component carries a drift term [Eq. 22] (lines 421–422), reduced for inertial particles by the local Stokes number $St = \tau_p/\tau_L$:

$$A_1 = A_3 = 0, \qquad A_2 = \frac{1}{1+St}\frac{\partial \sigma_2}{\partial y} = \frac{1}{1+St}\frac{u_\tau^2}{\nu}\frac{d(\sigma_2/u_\tau)}{dy^+}.$$

The derivative $d(\sigma_2/u_\tau)/dy^+$ is evaluated analytically (lines 163–188). The fluctuation is then $u'_i = \eta_i \sigma_i$ in the body-fitted frame and is rotated to Cartesian components.

5.4 Bulk branch ($y^+ \geq 100$)

Away from the wall the turbulence is treated as isotropic (lines 444–474), with $\sigma = (2k/3)^{1/2}$ [Eq. 26] and

$$\tau_L = \frac{2}{C_0}\frac{k}{\varepsilon}, \qquad C_0 = 14 \quad \text{[Eq. 36]}.$$

The bulk drift term of the model, proportional to ∇k [Eq. 27], is set to zero in this implementation (modification FIX-17 in the code). The code gives the reason that reading the cell gradient of k under the RSM caused a segmentation fault. The effect of omitting this term was not quantified.

5.5 Wall data and time step

The on-demand function `init_wall_udm` (lines 498–625) stores u_τ, y^+ and the wall normal in user-defined memory. It computes the wall shear stress from the tangential velocity of the wall-adjacent cell divided by the wall distance. The particle time step is $\Delta t = \max[\min(10^{-6}\ \text{s}, 0.1\,\tau_p), 10^{-9}\ \text{s}]$, capped by Fluent's own step (lines 765–797).

6. Results

6.1 Continuous phase

The RSM solution shows the secondary flow expected in a curved pipe: the centrifugal pressure gradient drives faster core fluid towards the outer wall and slower near-wall fluid inwards, forming a pair of counter-rotating Dean vortices. Revision 1 describes this flow qualitatively; no secondary-flow contours or profiles are included in either revision.

6.2 Deposition fraction

Table 1 lists the deposition fractions taken from the Fluent DPM summaries reproduced in Appendix B of revision 1 (also in `data/deposition_fraction.csv`). The Stokes number is $Stk = \rho_p d_p^2 U/(9\mu D)$ with the values of Section 2, neglecting the Cunningham correction.

Table 1. Deposition fraction (DF) per particle diameter.

d_p (µm)	Stk	Tracked	Escaped	Trapped	DF (%)	Sampling standard error (%)
10	0.092	10,000	8,329	1,671	16.7	0.37
20	0.368	10,000	4,279	5,721	57.2	0.49
30	0.829	10,000	2,142	7,858	78.6	0.41
40	1.473	10,000	546	9,454	94.5	0.23
50	2.302	10,000	199	9,801	98.0	0.14

The standard error, $\sqrt{DF(1-DF)/N}$, covers only the finite number of particles. It does not include the effect of the random-number issue described in Section 7.2.

Revision 1 (Fig. 10, kept as `figures/original-figure10-deposition.png`) also shows points at 5 µm (about 5 %) and 15 µm (about 33 %). No DPM summary is available for these two diameters, so they are not included in Table 1.

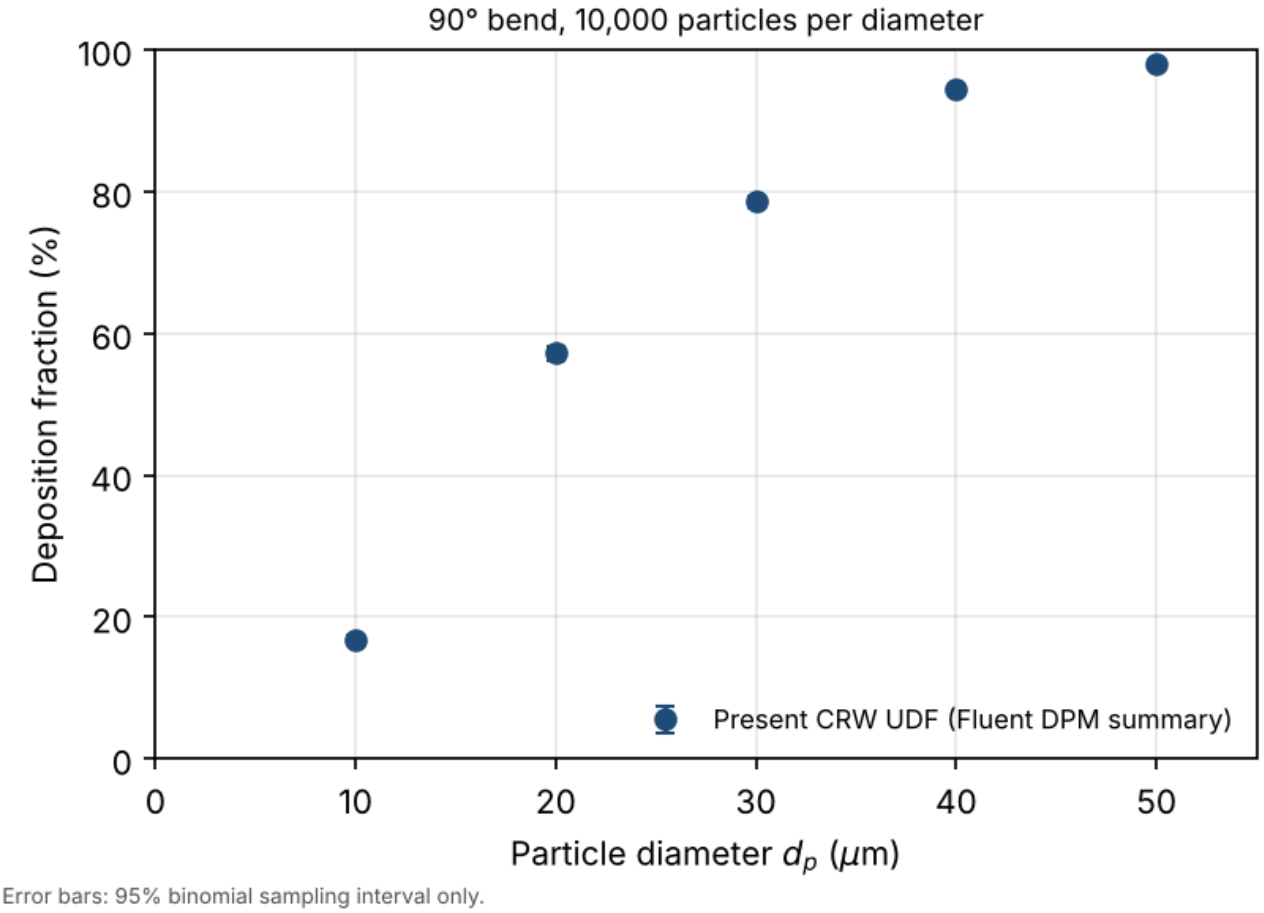


Figure 5. Deposition fraction from Table 1, generated by `tools/plot_deposition.py`.

6.3 Comparison with the benchmark

Revision 1 plots the simulated points together with a solid curve labelled “CRW Model”, described in the text as the numerical curve of Dehbi (2008). The curve passes through every simulated point, including changes of slope at 15 and 20 µm, and no experimental points of Pui et al. (1987) are shown. The figure therefore does not allow the reader to judge the agreement independently, and the statement “excellent agreement” is withdrawn until the following comparison is made:

1. Digitise Dehbi (2008, Fig. 8) and the Pui et al. (1987) data, recording the digitisation uncertainty.
2. Plot them against the values in Table 1, at matched Stokes number if the flow conditions differ (see the open item in Section 2).
3. Report the difference at each diameter.

7. Code review of the UDF

The source file was reviewed line by line for this revision. The file in this repository is unchanged from the version used for the results above.

7.1 The near-wall branch is applied only in wall-adjacent cells

`init_wall_udm` loops over wall faces and writes wall data only into the cell adjacent to each face (`F_C0`, lines 522–618). Every other cell keeps the flag `UDM_INIT = 0`. In `advance_langevin`, such cells fall back to the bulk branch:

lines 376–383 compute a y^+ from the wall distance and then overwrite it with `BL_YPLUS_MAX + 1`.

As a result, the anisotropic near-wall statistics and the drift correction of Section 5.3 act only in the first cell layer, where the cell-centre $y^+ \approx 0.5$. With 30 inflation layers, the cells between $y^+ \approx 1$ and $y^+ = 100$ use the isotropic, drift-free bulk branch, which is the treatment the Dehbi model was designed to replace. The reported agreement therefore cannot be attributed to the drift correction as implemented. The observed sensitivity to near-wall refinement (Section 3) may also be partly caused by this behaviour, because refinement changes the thickness of the single layer in which the near-wall branch is active.

Recommended correction: store u_τ, y^+ and the wall normal for all cells with $y^+ \le 100$, for example by assigning each cell the data of its nearest wall face and using `C_WALL_DIST` for the distance, then re-run all diameters.

7.2 The random-number generator produces correlated deviates

`gauss_rand` (lines 94–121) does not keep a random-number state. It hashes the particle's memory address, the current particle time and a salt through one step of two linear congruential generators and applies the Box–Muller transform. The program `tools/rng_check.c` runs a copy of this function with 32-bit `unsigned long`, as on the MSVC compiler used for the build. For a time step of 10^{-6} s and 2×10^5 samples it gives:

Statistic	Ideal value	Result
Mean, variance	0, 1	+0.001, 1.003
Autocorrelation, lags 1 to 5	0 ± 0.005	+0.22, +0.06, −0.11, −0.29, −0.23
Correlation between two particles at the same time (addresses 0x1C0 bytes apart)	0 ± 0.005	+0.13

The one-point statistics are correct, but successive deviates for the same particle are strongly correlated, and particles stored close together in memory receive correlated forcing. Serial correlation in ξ changes the effective diffusivity of the Langevin process, so the dispersion of particles is not the one the model prescribes.

Recommended correction: give each particle its own generator state, for example a 32-bit xorshift or SplitMix state stored in a particle user scalar and seeded once at injection, and re-run `rng_check` on the new generator before re-running the simulations.

7.3 Smaller points

1. The deposition comment (lines 741–752) states that a particle reaching a wall face is within one radius of the wall. Fluent detects wall contact when the particle centre reaches the face, so the trap is applied at zero distance, not at one radius. The difference matters most for the largest particles near the wall.
2. The bulk-branch comment states that omitting the ∇k drift has a "negligible effect on the final results". This was not tested.
3. In the fallback branch (lines 378–380) the computed y^+ is immediately overwritten; the two lines have no effect.
4. The particle Reynolds number in the drag factor uses only the first `ND_ND` components (3 in this 3D case) and the mean slip velocity, which is the usual approximation.

The code is otherwise well organised: it uses named constants for memory indices, guards against unassigned cells at injection, follows C89 declaration rules required by the Fluent compiler, and records each modification (FIX-2 to FIX-19) in comments.

8. Conclusions

1. A CRW–Langevin dispersion model following Dehbi (2008) was implemented as a compiled ANSYS Fluent UDF with anisotropic near-wall statistics, a body-fitted frame, a Stokes-number-weighted wall-normal drift and an adaptive particle time step.

2. On an RSM flow field with wall $y^+ \approx 0.95$, the UDF gives deposition fractions that rise from 16.7 % at 10 µm to 98.0 % at 50 µm (Table 1).
3. Deposition of 5 and 10 µm particles was sensitive to the near-wall mesh; this sensitivity was observed but not quantified.
4. The code review shows that, in the version used, the near-wall branch acts only in wall-adjacent cells and the random deviates are correlated. Until both are corrected and the benchmark comparison of Section 6.3 is made, the results should be described as a working implementation of the model with preliminary validation, not as a validated reproduction.

9. Use of software tools

As stated in revision 1, the Cursor code editor was used during UDF development to debug C syntax, memory-access errors and compilation problems with Fluent's `udf.h` macros, and to tidy comments. The physical model, equations and coupling logic were taken from Dehbi (2008).

References

1. Dehbi, A. (2008). Turbulent particle dispersion in arbitrary wall-bounded geometries: A coupled CFD-Langevin-equation based approach. *International Journal of Multiphase Flow*, 34(9), 819–828. https://doi.org/10.1016/j.ijmultiphaseflow.2008.03.001
2. Pui, D. Y. H., Romay-Novas, F., & Liu, B. Y. H. (1987). Experimental study of particle deposition in bends of circular cross section. *Aerosol Science and Technology*, 7(3), 301–315.
3. Dreeben, T. D., & Pope, S. B. (1997). Probability density function and Reynolds-stress modeling of near-wall turbulent flows. *Physics of Fluids*, 9(1), 154–163.
4. Kallio, G. A., & Reeks, M. W. (1989). A numerical simulation of particle deposition in turbulent boundary layers. *International Journal of Multiphase Flow*, 15(3), 433–446.